

Some theorems in favor of the guardrail just using $P(\text{Harm}|\text{action} = a)$

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This document discusses some results in favor of ‘naive conservatism’ — for any given situation, the guardrail just rejecting when $P(\text{Harm}|\text{action} = a) > \delta$. These results all roughly have the form ‘suppose we want the guardrail to have a certain property; then, its rejection rule should be $P(\text{Harm}|\text{action} = a) > \delta$ ’. (And naive conservatism indeed has these properties — the theorems are more like if-and-only-if’s.) In my opinion, each property is sort of compelling, but probably not individually sufficiently compelling to make it so that any individual theorem alone would provide sufficient reason to just settle on $P(\text{Harm}|\text{action} = a) > \delta$. However, I think these kinds of theorems together provide a decent amount of justification for guardrail rules vaguely like rejecting when $P(\text{Harm}|\text{action} = a) > \delta$ over other rules that look at the numbers $P(\text{Harm}|S_i, \text{action} = a)$ and $P(S_i)$ for some stories S_i .¹ So I don’t intend any one theorem as a killer argument in favor of looking at naive conservatism, but I think they jointly provide a significant amount of support to it. Minimally, you can view the theorems as pointing at some desiderata that other decision rules will fail to meet (but naive conservatism meets).²

¹More precisely, I think these theorems provide a decent amount of justification for basically just looking at $P(\text{Harm})$ if we’re looking at ‘rules like this’. I don’t think that a guardrail rejecting when $P(\text{Harm}) > \delta$ is anywhere close to the best decision rule if we look beyond decision rules involving various $P(\text{Harm})$ ’s, as I really think we should probably be doing various things differently: we should be collapsing the harm and goal specification into one, looking at expected utilities instead of $P(\text{Harm})$ ’s, treating inaction as just another action.

²I’ll also add the caveat that I just generally think something like $P(\text{Harm}|\text{action} = a) > \delta$ is better than other rules, and maybe you should be worried that I’m just being a ‘clever reasoner’ in favor of a predetermined conclusion (<https://www.lesswrong.com/posts/34XxbRFe54FycoCDw/the-bottom-line>). You might worry as follows: maybe I could also have listed some theorems in favor of a maximin decision rule, but chose to only list theorems in favor of the decision rule I favor? I think this is basically a prima facie fair concern, but I feel like the corpus of these kinds of theorems genuinely supports $P(H|a)$ -style decision rules (or, to my mind roughly equivalently (though I haven’t explained the connection here), expected utility maximization). But I guess me saying this should only allay the worry about me selectively presenting the results if you basically trust me to have a reasonable sense of the literature and to report this sense honestly — and it might be quite reasonable for you not to trust me on this (yet), in which case I might have trouble convincing you.

1 Setup

For various constructions, we will suppose that there is a background set of microstates $\mathcal{M}$. Each $M \in \mathcal{M}$ is an event, these events are a partition (they are mutually exclusive and collectively exhaustive), and for each action a , each microstate M determines with certainty whether that action leads to harm or not.³ We will be constructing various partitions of the sample space into macrostates out of microstates. This will mostly involve taking a union of the

1.1 The type signature of the guardrail

We will assume that the guardrail is given a partition $S_1, \dots, S_m$, together with $P(S_i)$ and $P(H|S_i, a)$,⁴ and it's supposed to output a binary decision — that is, whether the action is accepted or rejected. So, in general, we are allowing the guardrail's decision to depend on the partition. But we sometimes take there to be a fixed partition in the background, in which case we might as well think of the guardrail as just getting the list of pairs $((P(S_1), P(H|S_1)), \dots, (P(S_m), P(H|S_m)))$ as its input.

2 Partition-independence implies naive conservatism

In this section, we give a sequence of theorems, each of which (modulo some details) says that as long as we want the verdict of the guardrail (rejection vs not) to be independent of the chosen partition into $S_1, \dots, S_m$, the guardrail should just be checking if $P(H|a) < \delta$. Here's the same thing, elevated to an informal theorem:

Theorem 1 (Informal). *Suppose that the guardrail has the following two properties:*

Partition-invariance: The guardrail's decision cannot be changed by looking at the same action w.r.t. a different partition.

³We think of M as providing sufficient information about both the ethical theory and the world before the action to tell if the action causes harm or not. I guess it's possible to be a bit worried here about e.g. quantum randomness making this impossible or a sufficiently precise specification of the world also determining the action and then conditioning on some different action creating something weird and contradictory. The latter worry might be fixed if instead of conditioning on actions, we have a do operation instead. Anyway, I'll ignore this kind of worry for now, as it seems to me like it doesn't significantly reduce the force of the theorems — it seems plausible to me that alternative theorems could be stated in some setting where problematic assumptions are relaxed here, but I haven't thought about this carefully.

⁴This version of the story assumes that S_i and a are independent — otherwise, to compute $P(H|a)$, we'd want to have $P(S_i|a)$ and $P(H|S_i, a)$. Anyway, if S_i and a aren't independent, then we should probably just replace conditioning on a with do-ing a everywhere. We might get into some self-referential weirdness in this vicinity though.

Monotonicity: If action a is rejected⁵ and every single story⁶ thinks action a' is more harmful than a , then a' should also be rejected⁷.

Then, there is some δ such that the guardrail rejects iff $P(H) > \delta$.

There are various

3 Even for a fixed partition, a guardrail should aggregate

In this section, we will be assuming that actions can be probabilistically combined, yielding other actions — if you could do action a or action b , then for any $p \in [0, 1]$, you could also do the following action, which we denote $pa + (1 - p)b$:

- Flip a biased coin whose probability of landing on heads is p .⁸ Do a if it lands on heads, and b if it lands on tails.

It seems pretty reasonable to me that if the guardrail rejects a and also rejects b , it should reject the action that flips a coin and then does a or b depending on the result. If not, then the guardrail would be happy with an action before it learns some fact about the world, but whatever that fact turns out to be, it would definitely be unhappy with the action after learning that fact. Such a guardrail is ‘unstable over time’ — it lets through policies that it would for sure veto if just asked a bit later. To me, this seems quite irrational/undesirable. Similarly, it seems reasonable that if the guardrail accepts a and accepts b , then it should also accept $pa + (1 - p)b$ — if doing either of two things is considered fine, then surely choosing one of them to do at random and then doing it is also fine.

Theorem 2. *Suppose that the guardrail satisfies the following conditions:*

Sure-thing principle⁹ for rejection: If it rejects action a and rejects action b , then for any $p \in [0, 1]$, it should also reject the action $pa + (1 - p)b$.

Sure-thing principle for acceptance: If it accepts action a and accepts action b , then for any $p \in [0, 1]$, it should also accept the action $pa + (1 - p)b$.

⁵when presented in a particular partition

⁶in that same partition

⁷when presented in that particular partition

⁸In practice, if you have philosophical reasons to care about there in some sense being a determinate prior “truth of the matter” about the coinflip, you might want to replace the coinflip with some fact about the world that all theories assign probability $\approx p$, e.g., for $p = 1/2$, you could go with whether the number of protons in the Sun is currently odd or even.

4 More theorems!?

There is a sizable literature¹⁰ on various theorems in this general reference class in favor of averaging (this is just off the top of my head): Harsanyi's Utilitarian Theorem, the [von Neumann]–Morgenstern utility theorem, theorems by Jeffrey-Bolker and Savage, complete class theorems, Broome's philosophical work on aggregating value from different 'sites', and probably much more. These are often in terms of utility functions, but one can think of our 0/1 harm as just a 1/0 utility function, which maps probabilities of harm to expected utilities.

In particular, Harsanyi's theorem can be fairly straightforwardly translated into this setting. I started writing this up (the partial writeup is in the appendix), but then decided that what's now in Section 3 is probably (for our purposes) just a nicer version of the same (or at least a very similar) idea, making me think that a separate Harsanyi section beyond that is probably not worth writing.

Also, I could give a theorem that says that averaging is the most lenient rule satisfying a conservative max desideratum and an averaging desideratum, but this also started no longer feeling worthwhile to write up carefully in the wake of Section 3, as it does something really similar. Anyway, a version that I don't fully back is on Figure 1.

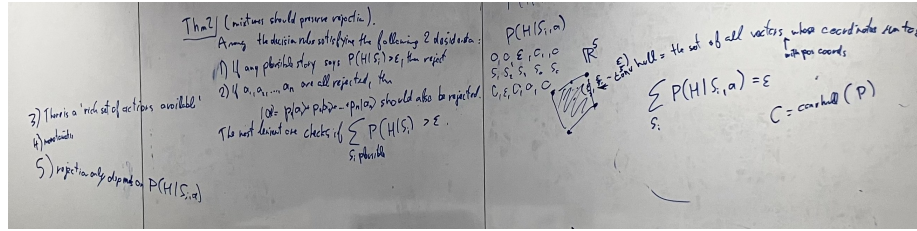


Figure 1: A theorem saying that among rules satisfying the conservatism supposed-desideratum, an imo reasonable substantive desideratum, and some technical conditions, averaging is the least conservative

¹⁰one could compile a sorta interesting reading list of like 100 papers, I'd guess

A Rational aggregation of opinions (uncouth notes)

In this section, we will assume a fixed reference partition $S_1, \dots, S_m$. For the next theorem, we will assume the guardrail has some internal structure — that it tracks actions being ‘less or more worthy of rejection’ in a reasonable way. (Think of this happening upstream of its actual rejection decision.) We will write $a \preceq b$ to denote the guardrail thinking a is at most as rejection-worthy as b .¹¹ We will assume that $\preceq$ satisfies the following properties:

Completeness: For any actions a, b , we have that $a \preceq b$ or $b \preceq a$.

Transitivity: For any actions a, b, c , if $a \preceq b$ and $b \preceq c$, then $a \preceq c$.

Continuity: For any actions a, b, c , if $a \preceq b \preceq c$, then there is a $p \in [0, 1]$ such that $b \sim pa + (1 - p)c$.¹²

Independence: For any actions a, b, c and $p \in (0, 1]$, we have $a \preceq b$ if and only if $pa + (1 - p)c \preceq pb + (1 - p)c$.

Coupling: If the guardrail accepts a and rejects b , then it thinks $a \prec b$.¹³

Let’s wrap this in a definition:

Definition A.1. We say the guardrail is rational if there is a 2-place relation $\preceq$ on actions satisfying Completeness, Transitivity, Continuity, Independence, and Coupling.

Theorem 3 (Harsányi’s Utilitarian Theorem, 1956 [1]). *We fix a reference partition $S_1, \dots, S_m$; let $\preceq_i$ denote the preferences of $i \in I$. Suppose the set of agents taken as a unit also has preferences denoted $\preceq_I$. Suppose that the following three properties all hold:*

Individual Rationality: For each $i \in I$, $\preceq_i$ is vNM-rational.

Collective Rationality: Additionally, $\preceq_I$ is vNM-rational.

¹¹Also define $a \succeq b$ to mean $b \preceq a$, and define $a \sim b$ to mean $[a \preceq b \text{ and } b \preceq a]$. And define $a \prec b$ to mean $[a \preceq b \text{ but not } b \preceq a]$, and $a \succ b$ to mean $[a \succeq b \text{ but not } b \succeq a]$.

¹²By $pa + (1 - p)c$, we mean the action that first flips a biased coin with probability p of landing on heads, then performs a if it lands on heads, and c if it doesn’t.

¹³We don’t require the converse, and indeed it will generally not hold.

Pareto Optimality: ¹⁴¹⁵ If $\mu \sim_i \nu$ for every $i \in I$, then $\mu \sim_I \nu$.

Then by the vNM theorem, there is a utility function $u_i: \mathcal{A} \rightarrow \mathbb{R}$ describing the preferences of each $i \in I$, as well as a utility function u_I describing the preferences of the group. But what's more, there are coefficients $c_0, c_1, \dots, c_k$ such that $u_I = c_0 + \sum_{i \in I} c_i u_i$.¹⁶

Theorem 4 (Harsányi's Utilitarian Theorem, 1956 [1]). *Let $I = \{1, 2, \dots, m\}$ index a set of agents, each with preferences; let $\preceq_i$ denote the preferences of $i \in I$. Suppose the set of agents taken as a unit also has preferences denoted $\preceq_I$. Suppose that the following three properties all hold:*

Individual Rationality: For each $i \in I$, $\preceq_i$ is vNM-rational.

Collective Rationality: Additionally, $\preceq_I$ is vNM-rational.

¹⁴I will now explain why I call this condition Pareto Optimality. Note that it is implied by the property that if $\mu \succeq_i \nu$ for every $i \in I$, then $\mu \succeq_I \nu$. Given some additional assumptions (for instance that there is a pair of lotteries λ, κ such that $\lambda \succeq_i \kappa$ for all $i \in I$ with at least one strict inequality), this condition follows from the condition that if everyone weakly prefers μ to ν with at least one $i \in I$ strictly preferring μ to ν , then $\mu \succ_I \nu$. The reason to call this last condition Pareto Optimality is that if it were not the case that $\mu \succ_I \nu$, then the "collective policy" $\succ_I$ would not necessarily pick μ over ν when given the choice, and every $i \in I$ would weakly prefer, with one $i \in I$ strictly preferring, a collective policy which is the same except for definitely picking μ over ν . In other words, the collective policy would not be Pareto optimal. And conversely, if there is some other collective policy which every $i \in I$ would non-strictly prefer and some $i \in I$ would strictly prefer to $\succeq_I$ regardless of the choice between two lotteries faced, then there must be some μ, ν such that $\mu \succeq_i \nu$ for all $i \in I$, $\mu \succ_i \nu$ for some $i \in I$, and $\mu \preceq_I \nu$. (However, if one assumes some probability distribution on the set of choices between lotteries faced, then one can write down a more stringent Pareto condition. One can also think of this distinction as the given condition being Pareto Optimality for agents with Knightian uncertainty over the decision problem that is to be faced.) The initial condition that $[\forall i, \mu \sim_i \nu] \implies \mu \sim_I \nu$ is not equivalent to this last Pareto condition, for instance because it is also satisfied when $u_I = -\sum_{i \in I} u_i$ which definitely does not in general satisfy the Pareto condition, but since it is (given some reasonable additional assumptions) implied by a genuine Pareto condition and conceptually not that far from it, I think it still makes sense to call it Pareto Optimality.

¹⁵I think there are reasons to only accept this axiom when individual preferences are based in personal welfare only. See the counterexample provided in footnote 7 here: <https://www.lesswrong.com/posts/DdmScdukXBff5CbNS>.

¹⁶A fun aside: note that the special case with $I = \{1\}$ says that any two utility functions u, v representing preferences which are indifferent between the same pairs of lotteries are affine functions of each other, i.e. $v = a + bu$. In particular, any two utility functions representing the same preferences are thus affine functions of each other; in fact, it is easy to see that we can always choose $b > 0$ in this case, and that given $b > 0$, we indeed have that $a + bu$ represents the same preferences. In other words, the utility function for a given set of preferences is unique up to positive affine transformation.

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